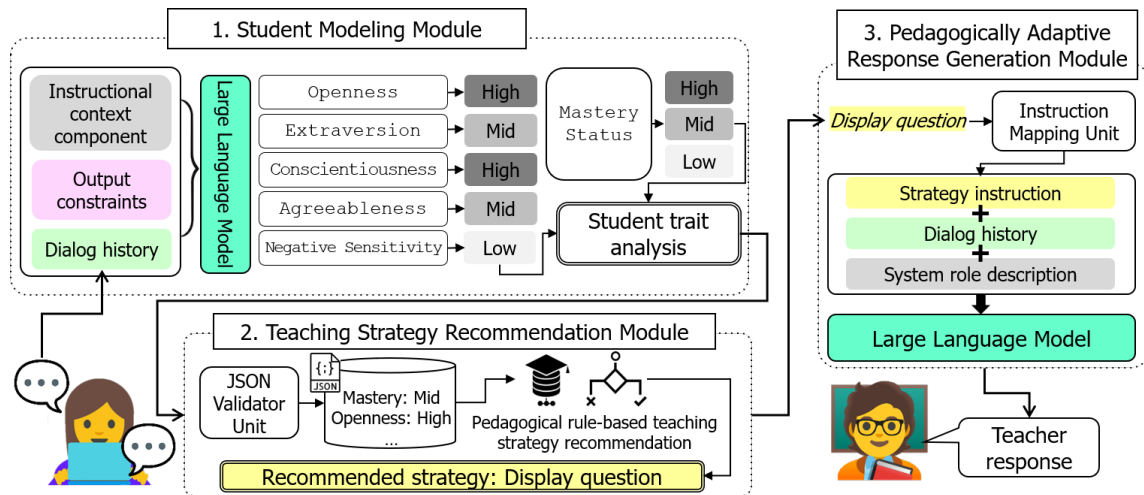


Graphical Abstract

Leveraging Large Language Models for Adaptive Tutoring System via Pedagogical Knowledge-Augmented Prompting



Highlights

Leveraging Large Language Models for Adaptive Tutoring System via Pedagogical Knowledge-Augmented Prompting

- Integrates student mastery status and Big Five-related behavioral tendencies into LLM-based tutoring for personalized instruction.
- Proposes a pedagogical knowledge-augmented prompting strategy that maps inferred learner profiles to specific teaching actions.
- Develops a rule-based pedagogical engine to guide LLM response generation based on the student's cognitive and non-cognitive traits.
- Achieves performance gain over baselines, validated through both automated metrics and qualitative evaluation of response quality.

Leveraging Large Language Models for Adaptive Tutoring System via Pedagogical Knowledge-Augmented Prompting

Abstract

Intelligent tutoring systems (ITSs) have gained increasing attention for their potential to support individualized instruction. While human tutors naturally adapt their teaching strategies by considering both cognitive and non-cognitive traits such as mastery level and personality, most existing ITSs lack mechanisms to model these individual differences effectively. This study proposes a framework for pedagogically adaptive tutoring that uses a large language model to infer students' mastery level and Big Five-related behavioral tendencies from dialogue history. The inferred traits guide teaching strategy selection through a rule-based pedagogical engine integrated into response generation. We evaluate the framework across six generative models including open-weight and proprietary systems using two public language-learning dialogue datasets (TSCC and CIMA). Performance is assessed through automatic metrics capturing lexical overlap (ROUGE-L), semantic similarity (BERTScore), and human-like engagement (DialogRPT), as well as a human evaluation study. Experimental results indicate consistent gains across models and datasets, with up to 6.7% relative improvement on automatic metrics and improved human-rated instructional quality. These findings suggest that integrating dialogue-based student modeling into response generation enhances the response-level instructional quality of automated tutoring.

Keywords: Intelligent Tutoring Systems, Large Language Models, Pedagogical Alignment, Student Modeling

1. Introduction

As digital learning environments become increasingly prevalent, the demand for scalable, personalized educational support continues to grow. Intelligent tutoring systems (ITS) have emerged as a powerful educational tool

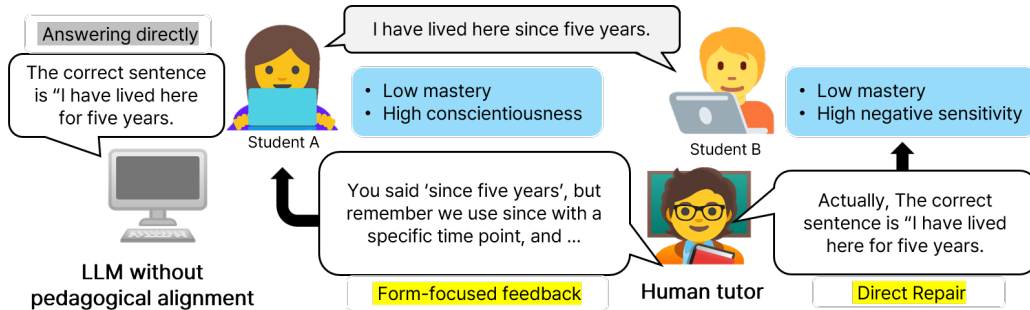


Figure 1: Example of human tutor’s adaptive feedback: The human tutor adjusts feedback type according to student traits.

by offering automated instruction to students through natural language conversations (Gaeta et al., 2025; Macina et al., 2023). These systems help students with diverse backgrounds achieve their educational goals with immediate feedback and flexible pacing, which traditional classroom settings often struggle to provide due to resource constraints.

Recent progress in large language models (LLMs) has accelerated research on dialog-based ITSs. A representative example is Khanmigo (KhanAcademy, 2024), a GPT-4-powered tutoring system developed by Khan Academy that engages students in personalized learning conversations. Earlier approaches focused on fine-tuning models on educational dialogue datasets (Hicke et al., 2023; Huber et al., 2023) or controlling generation through prompting techniques (Vasselli et al., 2023; Adigwe and Yuan, 2023; Omidvar and An, 2023; Piro et al., 2024). More recent work has explored structured prompting and interaction designs: Zhang et al. (2024) propose a playground-style interface encouraging guided reasoning, while Schmucker et al. (2024) analyze how conversational design choices affect engagement and instructional quality.

Despite progress in LLMs, replicating the adaptive strategies of human tutors remains challenging. As illustrated in Figure 1, human tutors adjust feedback based on learner characteristics. When a student makes an error (e.g., "I have lived here since five years"), a conscientious learner with low mastery receives form-focused feedback explaining the grammatical rule, while a sensitive learner at the same level receives direct correction to reduce confusion. In contrast, models like GPT-4 tend to default to giving direct answers regardless of learner needs (Borchers and Shou, 2025), lacking the trait-adaptive pedagogy that characterizes effective human tutoring.

Addressing this gap, recent ITS research highlights the need for peda-

gogically aligned LLM tutors capable of emulating human teachers’ dialog acts (Razafinirina et al., 2024). Earlier studies in this line have explored modeling instructional strategies for tutor-like response generation, such as teaching strategy prediction task (Lin et al., 2023; Suresh et al., 2019), and their integration into ITSs’ response generation (Wang et al., 2023) based on dialog history. However, these approaches often apply teaching strategies in the same way for all learners and do not consider how individual traits should influence those decisions. Personality modeling has been studied extensively using frameworks such as the Big Five (Goldberg, 1990), but limited research has explored a unified framework that dynamically links these traits to pedagogically adaptive LLM responses in real time.

To address this limitation, we present PATS (Pedagogically Adaptive Tutoring System), which integrates real-time student trait analysis with teaching strategy selection for pedagogically aligned LLM-based tutoring. We address this gap by introducing ‘pedagogical alignment’: constraining LLM generation through instructional strategies grounded in tutoring theory. We operationalize alignment through explicit rules connecting learner characteristics (mastery and behavioral tendencies) to specific teaching actions. Specifically, our system uses LLMs to infer key pedagogical traits—including the Big Five-related behavioral tendencies (openness, conscientiousness, extraversion, agreeableness, and negative sensitivity¹) and mastery levels—from the dialog history, then maps them to instructional strategies through a rule-based pedagogical engine.

Our contributions are threefold:

- We propose PATS, a framework that infers student traits from dialogue history and maps them to instructional strategies through a rule-based pedagogical engine.
- We demonstrate effectiveness across multiple LLM families and tutoring datasets, achieving 6.7% relative improvements in automatic metrics and significant gains in human-evaluated pedagogical quality.
- We provide validation of trait inference reliability and causal controls addressing potential confounds, supporting the interpretability of the

¹Throughout this paper, Negative Sensitivity refers to the trait commonly labeled Neuroticism in the standard Big Five taxonomy, relabeled here to reflect its manifestation in educational dialogue contexts better.

proposed framework.

To guide our evaluation, we address the following research questions:

RQ1: Can student mastery levels and Big Five-related behavioral tendencies be reliably inferred from dialogue history in an ITS setting?

RQ2: Does integrating trait-aware strategy selection improve the pedagogical quality of LLM-generated tutoring responses compared to baselines?

RQ3: Are the observed improvements attributable to structured pedagogical routing instead of the length of the prompts or just having strategy labels?

The rest of this paper is structured as follows. Section 2 provides an overview of related studies. Section 3 describes the proposed method, PATS. In Section 4, the proposed system is applied to language models for benchmarking, followed by an analysis and discussion of the experimental results. Section 5 concludes the paper.

2. Related Work

ITSs use natural language to provide personalized instruction, adapting to student needs by offering hints, explanations, or prompting questions. Studies show that such dialogue-based systems can effectively support skill development, especially in language learning (Bibauw et al., 2022).

Early ITSs such as AutoTutor (Graesser et al., 2004) relied on handcrafted scripts or decision trees, using expert-designed domain models to select tutor moves. These systems followed fixed frameworks (e.g., asking questions, evaluating answers, providing hints) and offered precise scaffolding. However, they required extensive rule engineering and lacked flexibility, limiting scalability in online contexts.

Early approaches in ITS leveraged machine learning sequence models like LSTMs and general-purpose Transformers to model tutoring dialogue. As language models advanced, pretrained encoder-decoder models were adapted through fine-tuning on educational corpora (Macina et al., 2023). However, these data-driven models, while improving dialog fluency, showed structural limitations related to their architectures and fine-tuning methods. Sequence-based models often do not capture long-range dependencies, causing inconsistent tracking of student progress across extended interactions (Razafinirina et al., 2024). Furthermore, fine-tuning on limited educational data restricts the diversity of learned teaching behaviors; without explicit integration of

pedagogical knowledge, the models tend to rely on their implicit pattern matching, leading to non-personalized responses.

More recently, LLMs are primarily adjusted for ITS through task-specific prompting, external retrieval, or minimal fine-tuning (Vasselli et al., 2023) with adapting methods like LoRA (Hu et al., 2021). This has led to various LLM control strategies: For instance, Vasselli et al. (2023) applied prompt ensembles with zero-shot and few-shot prompting, utilizing handcrafted and generated examples to improve LLM-generated teacher responses. They also used DialogRPT (Gao et al., 2020) as a metric to rerank candidate responses. Other researchers, like Sonkar et al. (2024), focused on modifying the LLM fine-tuning process by including features related to expected response sentiment (e.g., curiosity, approval) and response format to control the style and structure of the generated teacher utterance. Recent work further assessed LLMs’ instructional adaptivity by prompting with few-shot examples of tutor responses, finding that LLMs generally failed to reliably adjust feedback based on pedagogical context, often defaulting to direct corrections (Borchers and Shou, 2025). Despite these advancements, achieving adaptive tutoring remains challenging since these approaches do not consider the pedagogical decision-making process into response generation, lacking the mechanism to actively select the most effective instructional strategy for a given student.

To address this gap, two distinct lines of research have emerged. The first focuses on developing teaching strategy prediction modules. For example, Ganesh et al. (2021) trained a task-specific strategy classifier, while Wang et al. (2023) proposed a joint model for strategy prediction and response generation with self-distillation. Extending this line of research, Ikram et al. (2025) examined the performance of Llama 3 and GPT-4o to infer future tutor dialog acts in math tutoring scenario. However, these works mainly concentrate on improving the strategy prediction or the policy itself, focusing on predicting the next best action or following predefined scenario instructions. They generally fall short in incorporating student-specific mastery levels and behavioral tendencies into the decision-making process.

The second line of research emphasizes student modeling to improve ITS’s adaptivity. Sottolare et al. (2013) argued that mature ITSs require hierarchical student models capable of computing pedagogically relevant properties from interaction data. Katz et al. (2021) integrated students’ level of understanding modeling into physics tutorials, dynamically selecting questions. Lu et al. (2024) applied knowledge tracing to align responses with evolving student states, and Scarlatos et al. (2025) trained open-source LLMs to generate

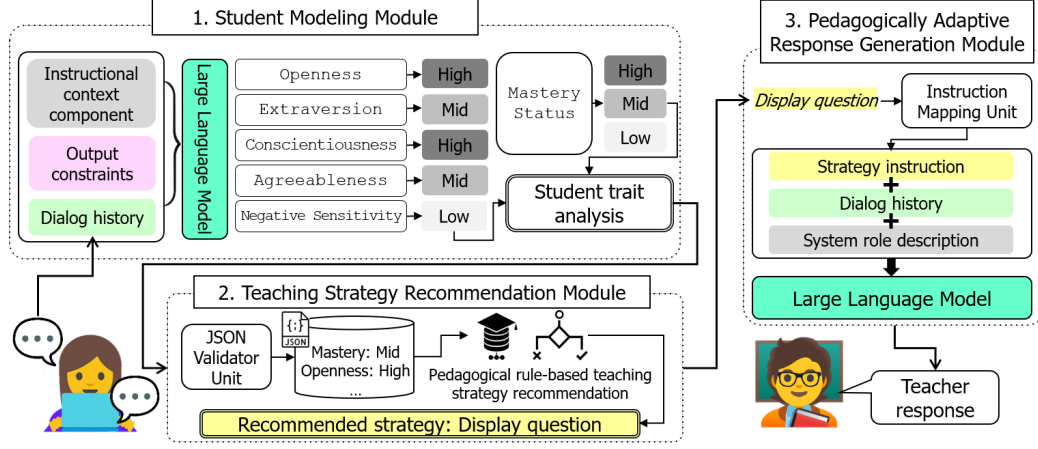


Figure 2: Overview of the proposed PATS framework.

tutor utterances that maximize the likelihood of student correctness. However, the predominant focus within this line of work is on cognitive states such as knowledge or mastery level, with little attention to non-cognitive traits like the Big Five-related behavioral tendencies that could directly modulate instructional delivery. This limitation is nontrivial, as prior research suggests that learners’ dispositions substantially affect the effectiveness of instructional methods (Trown, 1970; Bidjerano and Dai, 2007).

Collectively, a primary limitation in prior work is the disconnection between dynamic student modeling and the selection of instructional strategies. Our work addresses this gap by establishing an inferential loop where behavioral tendencies and mastery drive instructional decisions in real time.

3. Proposed Method

Our goal is to develop PATS, a framework for pedagogically adaptive tutoring that adjusts instructional strategies based on a student’s behavioral tendencies and mastery combination inferred from their dialog history. To achieve this, we define a three-stage framework that allows the ITS to conduct:

1. Student modeling: estimate the student’s mastery level and behavioral tendencies based on dialogue history.
2. Teaching strategy recommendation: select an appropriate instructional strategy based on the student’s inferred learning profile.

3. Pedagogically adaptive response generation: generate a tutoring response that incorporates the recommended instructional strategy.

We show an overall operation process of the PATS framework in Figure 2. The process begins with the student modeling module. Taking dialog history as input, this module uses LLM to analyze the student’s mastery status and Big Five-related behavioral tendencies and outputs the student profile in a JSON format. This profile is then passed to the teaching strategy recommendation module, where a rule-based pedagogical engine selects the most appropriate instructional strategy based on the specific trait combination and outputs the selected teaching strategy. Finally, the pedagogically adaptive response generation module takes the teaching strategy and conversational history as input. This module incorporates the chosen strategy into a structured prompt and guides the LLM to generate a tutor response as a final output.

3.1. Student Modeling

Behavioral tendencies such as openness and conscientiousness have been shown to shape how students engage with generative AI tools for learning (Arpaci et al., 2025), and recent learning analytics research argues that such stable individual differences should be treated as essential design resources in adaptive systems (Borchers et al., 2026). For instance, introverted students tend to prefer structured, rule-based instruction, while extraverted students respond better to interactive, example-driven approaches (Trown, 1970). Similarly, students with higher mastery benefit from more challenging tasks, whereas those with lower mastery often need step-by-step guidance (Bidjerano and Dai, 2007; Poropat, 2009; Komarraju et al., 2011). Accordingly, the student modeling module infers both mastery level and behavioral tendencies from dialogue history: mastery captures the cognitive dimension of the learner state, while behavioral tendencies reflect non-cognitive dispositions that modulate how instructional strategies are received and enacted.

The module adopts the Big Five personality model (Goldberg, 1990), widely used in educational research to represent individual differences across five dimensions. Unlike knowledge tracing approaches, which model cognitive states from structured item-response logs, our setting involves free-form dialogue and requires simultaneous inference of both mastery and behavioral tendencies; LLM-based inference is therefore adopted as a unified approach.

We define a tutoring dialogue d as a sequence of alternating student and tutor turns:

$$d = (s_1, t_1, s_2, \dots, t_{T-1}, s_T) \quad (1)$$

where s_k and t_k denote the student and tutor turns at timestep k , respectively. At any given timestep j , the dialogue history H_j captures the cumulative context up to the current student utterance:

$$H_j = (s_1, t_1, \dots, t_{j-1}, s_j) \quad (2)$$

Table 1 provides excerpts from the CIMA (Stasaski et al., 2020) and TSCC (Caines et al., 2022) datasets. Due to the brevity of CIMA dialogues (avg. 4.55 turns), the entire history is used for each sample. For TSCC, which has much longer dialogues, averaging 68.28 turns, we limit the input to the ten most recent turns. This keeps the context length manageable while still providing enough educational context.

The student modeling module infers a learning profile τ_j directly from the raw dialogue history H_j . This inference is operationalized via a large language model (GPT-3.5-turbo) conditioned on H_j and a specialized prompt P_{trait} :

$$\tau_j = \text{LLM}(P_{trait}, H_j) \quad (3)$$

The prompt P_{trait} assigns the LLM the persona of an educational psychologist, provides operational definitions for student traits, and instructs it to

Dataset	Excerpt Example
CIMA	Dialogue History: Teacher: "black" is "blu" in Italian. Student: But what are the other words? Teacher: Can you give me your best guess? Student: es en front de blu tree. Reference Response: Teacher: Getting there. Remember that the adjective always follows the noun it modifies.
TSCC	Dialogue History: Student: Whisky is made in Scotland. Teacher: good — only in the past or in general? Student: In general we can say that whisky is made in Scotland. Teacher: OK — thanks. Singular or plural? I mean, is 'whisky' countable? Student: I am not sure... maybe uncountable. Reference Response: Teacher: Yes, that's right... so we have to say: "Whisky ___ made in Scotland"?

Table 1: Excerpt examples from the CIMA and TSCC datasets showing dialogue histories and reference tutor responses.

directly output one of three categorical labels $\mathcal{L} = \{\text{High, Mid, Low}\}$ in a standardized JSON format. The full prompt is provided in Appendix A.1. The resulting profile τ_j is a six-dimensional vector representing both cognitive and non-cognitive states:

$$\tau_j = (M_j, O_j, E_j, C_j, A_j, N_j) \in \mathcal{L}^6 \quad (4)$$

where M_j denotes the student’s mastery level, and the remaining components represent the Big Five-related behavioral tendencies: Openness (O_j), Extraversion (E_j), Conscientiousness (C_j), Agreeableness (A_j), and Negative Sensitivity (N_j). Each dimension of the personality profile serves a distinct pedagogical role in our adaptive framework.

Openness (O_j) reflects intellectual curiosity. Students high in openness may respond well to exploratory questions or extended discussion, whereas those lower in openness may prefer more structured guidance (Jensen, 2015). Extraversion (E_j) relates to sociability and verbal expressiveness, which can be associated with a student’s level of active participation and verbal engagement during the dialogue (Rani Bhattacharjee and Ramkumar, 2025). For instance, more extraverted students may engage readily in open dialogue, whereas less extraverted students might respond more effectively to focused, concise prompts. Conscientiousness (C_j) describes the student’s degree of self-discipline and goal-directed behavior; it has been identified as a robust predictor of academic performance, matching or even exceeding the predictive power of cognitive ability in determining educational outcomes (Mammadov, 2022). Highly conscientious learners may benefit from clear task progression and explicit performance standards, while less conscientious students may require additional structure and follow-up prompts. Agreeableness (A_j) involves interpersonal cooperation and responsiveness to feedback. Learners high in agreeableness may accept corrective feedback more easily, while lower agreeableness may require more carefully framed guidance (Rani Bhattacharjee and Ramkumar, 2025). Finally, Negative Sensitivity (N_j) reflects a tendency toward anxiety or emotional instability, which can impede a student’s ability to handle academic stress and maintain resilience during challenging tasks (Mammadov, 2022).

Once τ_j is generated in JSON format, it undergoes a validation process to check conformity before being passed to the strategy recommendation module. To verify the reliability of our LLM-based trait inference, we compare these automated predictions against human-annotated subsets in Sec-

tion 4.6.2 to quantitatively measure the system’s psychological and pedagogical accuracy.

3.2. Teaching Strategy Recommendation Module

The teaching strategy recommendation module selects an appropriate instructional approach by mapping the student’s learning profile τ_j to a set of predefined strategies. We adopt a rule-based framework for this mapping to maintain transparency and keep the link between inferred traits and pedagogical actions grounded in educational theory. To determine the instructional strategy ξ_j , the system applies a mapping function f_{TS} :

$$\xi_j = f_{TS}(M_j, B_j) \quad (5)$$

where ξ_j denotes the strategy selected for turn j based on the joint consideration of the student’s current proficiency (M_j) and their behavioral disposition (B_j). By utilizing both cognitive and tendency-based dimensions, the function f_{TS} seeks to optimize the instruction for the specific learner profile. Our framework incorporates nine distinct teaching strategies derived from the Self-Evaluation of Teacher Talk (SETT) framework (Walsh, 2006). The framework offers a contextual perspective where language use acts as the interface between pedagogy and interaction, allowing the teacher’s instructional goals to be manifested through their verbal communication. The mapping function f_{TS} operationalizes established pedagogical principles by aligning specific instructional techniques with the student’s profile. Detailed rules defining this function are presented in Table 3, along with the pedagogical rationale for each strategy. Each rule condition is grounded in empirical findings on instructional effectiveness, accounting for the interplay

Teaching Strategy	Description
Direct Repair	Correcting an error quickly and directly
Display Question	Asking questions to which the teacher knows the answer
Referential Question	Genuine questions to which the teacher does not know the answer
Form-focused Feedback	Giving feedback on the words used, not the message
Seeking Clarification	Asking a student to clarify something the student has said
Extended Teacher Turn	Teacher turn of more than one clause
Scaffolding: Presentation	Explaining a language point
Scaffolding: Modeling	Providing an example of the target language feature
Scaffolding: Extension	Implicitly or explicitly encouraging more output

Table 2: Teaching strategy interactures derived from the SETT framework(Walsh, 2006; Caines et al., 2022)

Teaching Strategy	Rule Condition (Student Mastery + Behavioral Tendencies)	Pedagogical Rationale
Direct Repair	Student with <i>low mastery</i> and <i>mid to high negative sensitivity</i>	Immediate correction reduces cognitive overload for low-proficiency learners (Lin and Hedgcock, 1996; Hattie and Timperley, 2007). <i>"Actually, the correct answer is... Let's try it together."</i>
Display Question	Student with <i>low to medium mastery</i> and <i>high conscientiousness</i>	Structured recall strengthens memory and gap awareness in conscientious learners (Ellis, 1997; Shomoossi, 1997). <i>"When would you use a semicolon?"</i>
Referential Question	Student with <i>high openness</i> and <i>medium to high mastery</i>	Open learners with moderate mastery benefit from critical thinking and deeper engagement (Brock, 1986; Chaiklin et al., 2003). <i>"What would you do if you were in her place?"</i>
Form-focused Feedback	Student with <i>low to medium mastery</i> and <i>high conscientiousness</i> , when a specific linguistic error is detected.	Precise correction enhances accuracy for conscientious novices who value linguistic detail (Fotos and Nassaji, 2013). <i>"You said 'he go', but it should be 'he goes'. Do you remember the -s ending?"</i>
Seeking Clarification	Student with <i>medium mastery</i> and <i>low extraversion</i>	Promotes verbalization and confidence in shy learners (Swain, 1995; Baradeyah et al., 2017). <i>"Could you explain what you mean by that a bit more?"</i>
Extended Teacher Turn	Student with <i>low mastery</i> and <i>low extraversion</i> and <i>high agreeableness</i>	Preference to teacher-led long speech (Walsh, 2003; Dennis et al., 2016). <i>"Let me walk you through the whole process step-by-step."</i>
Scaffolding: Presentation	Student with <i>low mastery</i> and <i>high agreeableness</i>	Structured guidance supports initial comprehension (Vygotsky and Cole, 1978). <i>"Let's see how the past perfect tense is formed: we use 'had' + past participle."</i>
Scaffolding: Modeling	Student with <i>low to medium mastery</i> and <i>low conscientiousness</i>	Demonstration improves task understanding for learners needing procedural clarity (Anderson, 1983; Ghafaili, 2003). <i>"Watch how I solve this problem first, then you can try."</i>
Scaffolding: Extension	Student with <i>high mastery</i> and <i>high openness</i>	Encourages elaborative rehearsal and deeper learning (Vygotsky and Cole, 1978). <i>"That's right — can you think of another example where this applies?"</i>

Table 3: Rule-based mapping from student traits and learning context to teaching strategies, with example tutor utterances.

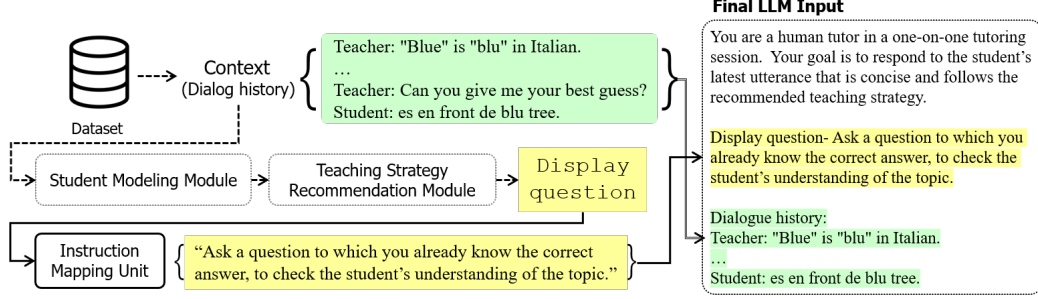


Figure 3: Internal flow detailing how the chosen strategy and context are converted into a structured instruction template for the LLM.

between mastery and behavioral tendencies. Strategy selection is jointly conditioned on both mastery and behavioral tendencies: mastery indicates the intensity of support required, while behavioral tendencies determine the interactional delivery. The system activates all strategy rules whose criteria are met, and when multiple strategies are triggered simultaneously, they are synthesized into a single response during generation. For example, when a student shows high mastery ($M_j = \text{High}$), the system selects strategies that encourage deeper application of knowledge, such as *Scaffolding: Extension* (Bloom, 1968). Conversely, for students with lower proficiency or higher anxiety, the mapping may favor more supportive or direct feedback modes. The selected strategy (or set of strategies) ξ_j is subsequently passed to the response generation module to guide the synthesis of the final tutor utterance.

3.3. Pedagogically Adaptive Response Generation Module

The pedagogically adaptive response generation module generates instructional responses based on the selected teaching strategy ξ_j and the student's dialog history. This module uses a generative language model to produce tutor replies conditioned on the specified pedagogical strategy while maintaining dialogue coherence. The module takes the teaching strategy ξ_j and the student's dialog history H_j up to turn j as input.

The system formulates a structured prompt to guide the LLM's output, embedding the pedagogical decision ξ_j —selected deterministically by the rule-based engine prior to generation—as a textual instruction. Figure 3 illustrates this process. The LLM-based response generation can be formally expressed as:

$$T_j = \text{LLM}(\text{SystemRole} || \text{Strategy}(\xi_j) || \text{Context}(H_j)) \quad (6)$$

where T_j is the generated tutor response. *SystemRole* defines the general tutor persona and global constraints, *Strategy* inserts explicit instructions corresponding to ξ_j , and *Context* includes the dialogue history H_j . The operator $||$ denotes string concatenation. The LLM thus operates as a conditioned generator, producing responses guided by the externally provided strategy block rather than selecting or inferring strategies during generation.

The instruction mapping unit converts each strategy label ξ_j into a written instruction set. For example, if the selected strategy is “Display question,” the prompt includes an instruction such as: “Ask a question to which you already know the correct answer to check the student’s understanding of the topic.” Additional mappings are shown in Table 4. To evaluate the extent to which the intended strategy is reflected in the generated outputs, we conducted a human validation study in Section 4.6.3.

4. Experiment

4.1. Datasets

In this section, we provide an overview of our datasets, experimental setup, and findings from using the proposed method. We evaluated our method using two publicly available language tutoring dialogue datasets: CIMA and TSCC. Both datasets contain conversations between tutors and students. CIMA (Stasaski et al., 2020) is a crowd-sourced dataset where annotators acted as students and teachers practicing translations of English prepositional phrases into Italian. TSCC (Caines et al., 2022) includes written one-on-one English tutoring sessions, capturing real exchanges between students and teachers. While CIMA follows a controlled, role-play setting, TSCC reflects more natural tutoring interactions and authentic teaching dynamics. Due to the difference in dialogue length between the two datasets (as described in Section 3.1), we use the full conversation history for CIMA,

Teaching Strategy	Prompt Template Example
Direct Repair	Provide the correct answer and explain the reasoning clearly.
Display Question	Ask a question to which you already know the correct answer, to check the student’s understanding of the topic.
Referential Question	Ask an open-ended question for which you do not know the answer, encouraging the student to share personal ideas or experiences.

Table 4: Example prompt instructions used to guide the LLM’s response generation based on the selected teaching strategy.

whereas for TSCC, the ten preceding turns before each strategy-annotated teacher utterance are included.

Dataset strategy annotations are excluded from model input; in all settings, the model receives only raw dialogue history, and strategy recommendations are generated solely by the rule-based module.

4.2. Models and Generation Settings

This study utilizes a total of six language models: three open-source models (T5-v1.1-large, LLaMA-3.2-1B, and Mistral 7B) and three proprietary models (GPT-3.5-Turbo, GPT-4, and Gemini 2.5 Flash). T5-v1.1-large (Raffel et al., 2020) features 770 million parameters and uses an encoder-decoder structure. LLaMA-3.2-1B (Grattafiori et al., 2024) is an auto-regressive model employing an optimized transformer architecture, and Mistral 7B (Jiang et al., 2023) is an auto-regressive language model with 7 billion parameters, known for its efficiency. All open-source models were accessed via the Hugging Face Transformers library. The proprietary models were evaluated using their respective APIs: OpenAI API for GPT models and Google AI Studio for the Gemini model.

All experiments involving open-source models were conducted on a single NVIDIA RTX 3060 GPU, which provides 12 GB of VRAM. The T5 and LLaMA-3.2-1B model was fine-tuned using the Low-Rank Adaptation(LoRA) (Hu et al., 2021) method. The LoRA adapter was configured with a rank (r) of 8 and an α of 16. The training utilized an AdamW optimizer and a batch size of 8. The Mistral 7B model was fine-tuned using the more memory-efficient Quantized Low-Rank Adaptation(Dettmers et al., 2023) method due to its 7-billion parameter size and the VRAM constraint. The LoRA parameters were set identically to the LLaMA experiment ($r = 8, \alpha = 16$) with a batch size of 1, combined with 8 steps of gradient accumulation. The inference and the evaluation of the API-based GPT and Gemini models were conducted without further fine-tuning. Trait inference used temperature=0.1 for stable categorical assignment, while response generation used temperature=0.7. For open-weight models, stochastic decoding was applied (do_sample=True, temperature=0.7, top_p=0.8). For API-based models (GPT-3.5, GPT-4, Gemini), generation used temperature=0.7 under default API settings. All experiments were repeated three times with identical parameters, and results are averaged.

4.3. Baseline and Proposed Configurations

Baseline Setting: w/o PATS In the "w/o PATS" setting, we evaluate the generation model without the integration of adapted teaching strategies. In this configuration, the model receives only basic context information, such as the student's previous conversation turns. The prompt used for testing in this setting is as follows:

You are a human tutor in a one-on-one tutoring session. Your goal is to respond to the student's latest utterance.

Dialogue history: [turn1] [turn2] ... [student_latest_utterance]

This prompt provides the model with the conversational context, but does not include any information about recommended teaching strategies.

Proposed method: PATS The "PATS" setting represents our proposed method, which incorporates teaching strategies recommended by the strategy recommendation module. In this setting, the model uses prompts that include both the dialogue history and the suggested teaching strategies. The prompt for this setting is as follows:

You are a human tutor in a one-on-one tutoring session. Your goal is to respond to the student's latest utterance that is concise and follows the recommended teaching strategy.

Recommended teaching strategy: [Strategy Name] [Strategy Description]

Dialogue history: [turn1] [turn2] ... [student_latest_utterance]

4.4. Evaluation Methodology

4.4.1. Automatic Metrics

We evaluated response quality using both reference-based and reference-free method. In all settings, the model generates a single tutor response conditioned on the dialogue history, without simulating future student turns. For reference-based methods, we used ROUGE-L(Lin, 2004) and BERTScore F1 (Zhang et al., 2019) to measure similarity between generated and reference teacher responses. ROUGE-L is a recall-based metric that measures the longest common subsequence word overlap between the generated and reference sentences. Since the reference responses are drawn from human tutors, higher lexical overlap with these references reflects closer approximation to expert teaching language. BERTScore F1 assesses semantic similarity based

on cosine similarity of token embeddings. The implementation followed HuggingFace’s evaluate package with the distilbert-base-uncased model.

We also applied DialogRPT (Gao et al., 2020), a set of reference-free neural evaluation models trained on millions of human feedback data. It assesses dialogue quality by predicting how likely the response is to receive engagement. For each model, the final BERTScore was the mean F1 across all test items, and the DialogRPT score was averaged across three metrics (updown, human vs. rand, human vs. machine). For stable evaluation, response generation and evaluation were conducted three times, with the final result reported as the average.

These metrics capture surface-level similarity and engagement rather than pedagogical correctness directly, and are therefore complemented by human assessment (Section 4.6.6) focusing on instructional quality.

4.4.2. Human Evaluation Protocol

Automatic metrics alone do not fully capture the pedagogical quality of tutoring responses. We therefore conducted a human evaluation to assess the instructional adequacy of the generated responses. We recruited ten experienced English teachers from middle and high schools in South Korea. The evaluation set consisted of 100 dialogue items. Each triplet included three responses to the same student utterance: (1) the expert-written reference response, (2) the response generated by PATS, and (3) the response generated without PATS (w/o PATS). Each triplet was evaluated by three annotators to ensure reliability, resulting in a total of 300 ratings. We used an overlapping assignment design: each annotator evaluated 30 items, and each item was rated by three different annotators.

Responses were assessed across four criteria grounded in prior research on educational dialogue systems (Upadhyay et al., 2023; Liu et al., 2025): (1) adaptivity, (2) engagement, (3) teacher-likeness, and (4) helpfulness. (Upadhyay et al., 2023; Liu et al., 2025)

4.4.3. Personality Inference Validation Protocol

To evaluate the reliability of dialogue-based personality inference, we sampled 50 dialogue excerpts from tutoring datasets and asked five annotators with backgrounds in educational psychology to rate each student’s Big Five traits on a three-level scale (high, medium, low). These ratings served as a proxy ground truth.

We first measured inter-annotator agreement using Fleiss’s κ , which assesses consistency among multiple raters when assigning categorical ratings or classifying items. Following the conventional interpretation scale (Landis and Koch, 1977), κ values between 0.01–0.20 indicate slight agreement, 0.21–0.40 fair, 0.41–0.60 moderate, 0.61–0.80 substantial, and above 0.81 almost perfect agreement. Then, we compared the LLM-inferred labels with the aggregated expert labels using Accuracy and Cohen’s κ . Accuracy shows how often the predictions match the expert labels, while Cohen’s κ provides a chance-corrected measure, offering a more reliable evaluation under class imbalance. According to standard interpretation guidelines (Landis and Koch, 1977), Cohen’s κ values below 0.40 indicate poor agreement, 0.41–0.60 moderate, 0.61–0.80 substantial, and above 0.81 almost perfect agreement.

4.5. Validation Experiments

4.5.1. Ablation Study Design

To isolate the contribution of each component in PATS, we construct several ablation variants.

Mastery-only In this setting, strategy selection is based only on the student’s mastery level. Behavioral tendencies are not used for routing.

Tendency-only In this setting, strategy selection is based only on inferred behavioral tendencies. Mastery information is not considered.

Gold Trait + Rule In this variant, inferred traits are replaced with human-annotated gold trait labels, while the same rule-based strategy mapping is applied. This setting assesses the effect of trait inference accuracy on downstream strategy selection.

Random Trait + Rule In this setting, inferred trait levels are replaced with randomly sampled levels from {Low, Mid, High} for each trait dimension. The rule-based routing remains unchanged. This simulates severe trait misclassification and allows us to examine error propagation.

No Template In this variant, the strategy is selected by the rule-based module, but the instruction template of the strategy is removed from the prompt. This tests the role of explicit strategy realization in generation.

4.5.2. Causal Control Conditions

Because PATS adds an additional strategy block in the prompt, performance gains may reflect increased prompt length or explicit instructions. To control for this possibility, we introduce two additional conditions.

Placebo (Length-Matched Control) In this setting, we retain the same prompt structure as PATS but replace the strategy block with a length-matched instruction that does not contain pedagogical guidance. The strategy section is replaced with a placeholder text that preserves the format and approximate length of the strategy block:

Recommended teaching strategy: [Response Guidelines]
 Use complete sentences with proper grammar and provide a response that addresses their input.

Random Strategy Selection (RSS) In this condition, a strategy label and description are randomly sampled from the predefined strategy set, independent of the student’s mastery level or inferred traits. The sampled strategy is inserted into the same prompt position as in PATS. This setting tests whether including a strategy label alone improves performance, regardless of its contextual relevance.

4.6. Experimental Results

4.6.1. Overall Performance Across Models

We present the results of our experiments under two settings in Table 5. The integration of PATS consistently improves model performance across both datasets. For example, the BERTScore of T5-large increases from 0.746 to 0.756 on CIMA and from 0.761 to 0.773 on TSCC. A similar trend is observed for LLaMA 3.2–1B, where the BERTScore improves from 0.730 to 0.739 on CIMA and from 0.782 to 0.799 on TSCC. The largest relative improvement (6.7%) was observed for GPT-3.5 on the CIMA dataset using Rouge-L, where the score increased from 40.3 to 43.0. These results indicate that providing explicit teaching strategy labels helps the models generate more pedagogically coherent responses. Among open-weight models, Mistral-7B achieves the best overall performance, showing higher Rouge-L, BERTScore, and DialogRPT values than LLaMA 3.2–1B and T5-large. Its BERTScore improves from 0.787 to 0.799 on CIMA and from 0.803 to 0.815 on TSCC, demonstrating improved performance even for larger open-source models. As expected, proprietary models such as GPT-3.5, GPT-4, and Gemini 2.5 Flash outperform open-weight models overall, with GPT-4 and Gemini 2.5 Flash achieving the highest BERTScores (up to 0.834 and 0.840, respectively). These differences reflect disparities in model scale and training resources.

The overall BERTScore improvement is clearer in TSCC than in CIMA. For example, GPT-4 shows an increase of 0.3pp on CIMA and 0.9pp on TSCC, with similar patterns appearing in Mistral-7B and LLaMA 3.2–1B. This stronger effect in TSCC is likely due to its wider range of dialog acts (23 types) compared to CIMA (5 types), suggesting that our framework’s fine-grained pedagogical strategies align well with the dataset’s semantic and instructional diversity. On the other hand, the higher ROUGE-L scores on CIMA likely reflect differences in how the datasets were created. CIMA was built through crowdsourced role-play, where annotators acted as both students and teachers, producing well-structured and grammatically consistent utterances that align with pretrained LLM language patterns. In contrast, TSCC consists of authentic tutoring chats, which include informal phrasing and incomplete sentences characteristic of spontaneous dialogue. These natural irregularities reduce surface-level lexical overlap between model outputs and references, resulting in lower ROUGE-L score improvements despite pedagogically sound responses in TSCC.

Across all models, DialogRPT gains are smaller than those of BERTScore.

Model	Metric	TSCC (w/o PATS)	TSCC (PATS)	CIMA (w/o PATS)	CIMA (PATS)
T5-large (770M)	Rouge-L	41.1 (± 0.3)	41.7 (± 0.3)	35.6 (± 0.4)	37.1 (± 0.5)*
	BERTScore	0.761 (± 0.002)	0.773 (± 0.002)	0.746 (± 0.006)	0.756 (± 0.005)*
	DialogRPT	0.43 (± 0.01)	0.45 (± 0.01)*	0.52 (± 0.02)	0.53 (± 0.01)
LLaMA 3.2 (1B)	Rouge-L	41.0 (± 0.3)	42.2 (± 0.2)*	38.1 (± 0.3)	38.8 (± 0.3)*
	BERTScore	0.782 (± 0.002)	0.799 (± 0.002)*	0.730 (± 0.003)	0.739 (± 0.006)*
	DialogRPT	0.47 (± 0.03)	0.49 (± 0.02)	0.53 (± 0.01)	0.55 (± 0.01)*
Mistral-7B	Rouge-L	43.2 (± 0.3)	44.6 (± 0.2)*	40.8 (± 0.3)	41.9 (± 0.3)*
	BERTScore	0.803 (± 0.002)	0.815 (± 0.002)*	0.787 (± 0.006)	0.799 (± 0.003)*
	DialogRPT	0.50 (± 0.01)	0.54 (± 0.01)*	0.51 (± 0.01)	0.54 (± 0.01)*
GPT-3.5	Rouge-L	41.8 (± 0.3)	42.0 (± 0.2)	40.3 (± 0.3)	43.0 (± 0.3)*
	BERTScore	0.815 (± 0.002)	0.823 (± 0.002)*	0.813 (± 0.002)	0.813 (± 0.002)
	DialogRPT	0.52 (± 0.01)	0.54 (± 0.02)	0.52 (± 0.02)	0.55 (± 0.01)*
GPT-4	Rouge-L	41.2 (± 0.3)	42.1 (± 0.2)*	40.5 (± 0.3)	42.0 (± 0.3)*
	BERTScore	0.825 (± 0.002)	0.834 (± 0.001)	0.822 (± 0.002)	0.825 (± 0.002)
	DialogRPT	0.48 (± 0.03)	0.50 (± 0.01)*	0.55 (± 0.01)	0.55 (± 0.01)
Gemini 2.5	Rouge-L	41.3 (± 0.3)	42.1 (± 0.2)*	40.9 (± 0.3)	41.1 (± 0.2)
Flash	BERTScore	0.823 (± 0.002)	0.840 (± 0.001)*	0.825 (± 0.002)	0.831 (± 0.002)*
	DialogRPT	0.48 (± 0.01)	0.51 (± 0.01)*	0.55 (± 0.01)	0.56 (± 0.01)

Table 5: Mean ($\pm$ SD) performance of models with and without PATS on CIMA and TSCC datasets, measured by Rouge-L, BERTScore, and DialogRPT. * indicates $p < 0.05$, Wilcoxon Signed-Rank Test).

This suggests that while PATS helps models generate more semantically consistent responses, it has a weaker impact on the perceived engagement or natural flow of conversation. In other words, PATS mainly focuses on how the model organizes explanations and delivers pedagogically coherent feedback, rather than changing tone or expressiveness, which are factors that DialogRPT measures.

4.6.2. Validation of Personality Inference

To ground our LLM-based student modeling approach, we conducted an evaluation comparing LLM-inferred traits against human expert annotations. The inter-annotator consistency among the five experts was quantified using Fleiss’ κ . Subsequently, the LLM-inferred labels for the same dialogue excerpts were generated. The LLM performance was evaluated against the majority-vote labels from the experts using two primary metrics: Accuracy and Cohen’s κ . The results, summarized in Table 6, show that the LLM-based model achieves a substantial level of agreement with human expert judgments. The expert annotations themselves showed a moderate inter-annotator agreement (Average Fleiss’ $\kappa = 0.69$), which reflects the inherently nuanced nature of inferring psychological traits from short dialogue segments rather than inconsistency among annotators. The LLM-inferred traits showed a solid overall performance, achieving an average accuracy of 0.73 and an Average Cohen’s κ of 0.64 against the expert majority labels. The model showed the highest performance in predicting mastery (accuracy: 0.82, Cohen’s κ : 0.72) and Openness (Accuracy: 0.72, Cohen’s κ : 0.65). This good performance suggests the LLM is effective at detecting explicit linguistic cues related to proficiency and intellectual curiosity from the dialogue history. Conversely, the model achieved the lowest agreement on Negative sensitivity (Cohen’s κ : 0.57) and Extraversion (Cohen’s κ : 0.60), which aligns with the lower inter-annotator agreement observed for these traits. This indicates that

Trait	Fleiss’ κ (Experts)	Accuracy (LLM vs Experts)	Cohen’s κ (LLM vs Experts)
Mastery	0.78	0.82	0.72
Openness	0.72	0.72	0.65
Conscientiousness	0.65	0.75	0.63
Extraversion	0.68	0.67	0.6
Agreeableness	0.74	0.73	0.67
Negative sensitivity	0.60	0.7	0.57
Average	0.69	0.73	0.64

Table 6: Agreement between expert annotations (Fleiss’ κ) and LLM-based inference performance (Accuracy, Cohen’s κ) across the Mastery and Big Five traits.

inferring these non-cognitive traits from text remains a nuanced challenge, both for human raters and the LLM.

To examine whether inferred traits reflect stable characteristics, we measured within-session consistency on TSCC. Since CIMA consists of simulated dialogues, we report stability results on TSCC only. Each session was segmented into 10-turn windows, and we computed the proportion of window pairs yielding the same trait level for each dimension. For a session with K windows and D trait dimensions, we compute pairwise agreement as:

$$A_s = \frac{1}{\binom{K}{2}} \sum_{i < j} \left(\frac{1}{D} \sum_{d=1}^D \mathbf{1}(T_{s,i}^{(d)} = T_{s,j}^{(d)}) \right).$$

where $T_{s,i}$ denotes the inferred trait level for window i in session s , and $\mathbf{1}(\cdot)$ is an indicator function that equals 1 if the two trait levels are the same and 0 otherwise. The mean agreement $\bar{A}$ across sessions was 76.3%, suggesting that inferred trait levels remain relatively consistent as dialogue progresses. These results suggest that inferred trait levels show moderate within-session stability. In practice, teachers typically work with broad performance categories rather than continuous proficiency scores, and the inferred labels are best understood as dialogue-observable behavioral tendencies in this spirit. The ablation results in Section 4.6.4 support this: when trait labels are replaced with randomly sampled levels, performance drops below the no-strategy baseline (e.g., DialogRPT 0.47 for Mistral-7B), indicating that directional accuracy in trait inference, not fine-grained precision, drives effective strategy selection.

4.6.3. Adaptive Strategy Selection

A potential concern with rule-based strategy selection is that the system may repeatedly select the same strategy. To examine this, we analyzed strategy distributions across 80 TSCC dialogue instances using GPT-4, comparing the w/o PATS and full PATS settings (160 responses total). Two annotators with certified middle-school teacher status independently labeled the realized

Setting	Entropy (H)	Top-1 Share
w/o PATS	0.96	0.79
PATS (Full)	1.13	0.56

Table 7: Strategy distribution statistics on the TSCC dataset ($N = 80$ responses per setting).

strategy for each response (Cohen’s $\kappa = 0.68$), with disagreements resolved through discussion.

We report two statistics: entropy of the strategy distribution and Top-1 strategy share. Under the w/o PATS setting, strategy usage is highly concentrated, with a Top-1 share of 79% and entropy of $H = 0.96$. PATS reduces this concentration, yielding a Top-1 share of 56% and entropy of $H = 1.13$, indicating more balanced strategy use across responses.

We additionally computed a strategy adherence rate: the proportion of responses in which the annotators identified the intended strategy. We found that 88.75% of PATS responses reflected the prompted strategy. To assess whether this diversity reflects pedagogically appropriate adaptation, we examined the alignment between strategy selection and learner profiles (§ 4.6.9). Results show that strategy distribution shifts meaningfully by mastery level and behavioral dimensions.

Model	Setting	Rouge-L	BERTScore	DialogRPT
Mistral-7B	w/o PATS	43.2 (± 0.3)	0.803 (± 0.002)	0.50 (± 0.01)
	Mastery-only	44.0 (± 0.3)	0.812 (± 0.002)	0.51 (± 0.01)
	Tendency-only	43.3 (± 0.3)	0.805 (± 0.002)	0.53 (± 0.01)
	Gold Trait + Rule	44.1 (± 0.2)	0.820 (± 0.001)	0.53 (± 0.01)
	Random Trait + Rule	42.3 (± 0.4)	0.798 (± 0.003)	0.47 (± 0.02)
	No Template	43.9 (± 0.3)	0.807 (± 0.002)	0.49 (± 0.01)
	PATS (Full)	$\dagger 44.6 (\pm 0.2)$	$\dagger 0.815 (\pm 0.002)$	$\dagger 0.54 (\pm 0.01)$
GPT-4	w/o PATS	41.2 (± 0.3)	0.825 (± 0.002)	0.48 (± 0.03)
	Mastery-only	41.7 (± 0.3)	0.831 (± 0.002)	0.48 (± 0.02)
	Tendency-only	41.6 (± 0.3)	0.827 (± 0.002)	0.5 (± 0.02)
	Gold Trait + Rule	42.2 (± 0.2)	0.836 (± 0.001)	0.51 (± 0.01)
	Random Trait + Rule	40.8 (± 0.4)	0.827 (± 0.003)	0.46 (± 0.02)
	No Template	41.6 (± 0.3)	0.828 (± 0.002)	0.47 (± 0.02)
	PATS (Full)	42.1 (± 0.2)	$\dagger 0.834 (\pm 0.001)$	$\dagger 0.50 (\pm 0.01)$

Table 8: Module-level ablation study isolating the contribution of consisting modules. Bold denotes the best performance among automatically-inferred settings (excluding Gold Trait + Rule) and $\dagger$ indicates statistical significance ($p < 0.05$) compared to the Mastery-only, Tendency-only, and Random variants (Wilcoxon signed-rank test).

4.6.4. Module Ablation Analysis

To examine the contribution of each component, we compare the full PATS model against five variants: w/o PATS, mastery-only routing, tendency-only routing, gold trait labels with rule-based routing, random trait labels with rule-based routing, and strategy selection without instruction templates.

Table 8 presents the results. Removing either mastery or tendency routing degrades performance, with neither partial configuration matching the full model. Mastery signals show a more pronounced effect on text-overlap metrics such as ROUGE-L, while tendency-based routing contributes more to engagement scores such as DialogRPT, suggesting that cognitive and non-cognitive signals play complementary roles.

The Random Trait variant yields the lowest performance across all metrics, falling below the w/o PATS baseline (e.g., DialogRPT drops to 0.47 for Mistral-7B), confirming that misaligned strategy selection degrades response quality. The marginal gap between Gold Trait and full PATS (BERTScore difference of 0.005 for Mistral-7B) suggests that LLM-based trait inference is sufficiently reliable for pedagogical routing. Finally, removing instruction templates while retaining strategy selection also reduces performance, indicating that explicit prompt structure is necessary for the LLM to realize the intended pedagogical move.

4.6.5. Causal Controls for Confounds

To evaluate whether performance gains stem from pedagogical alignment, we compare PATS against the placebo and RSS settings described in Section 4.5.2. Table 9 presents the results across three model families. The placebo condition shows minimal deviation from the baseline despite the increased instruction length. In contrast, the RSS often leads to lower performance; Mistral-7B ROUGE-L drops to 41.4, suggesting that misaligned strategy instructions can interfere with response quality. PATS consistently yields the highest scores across all metrics. These results suggest that the observed improvements are not an artifact of prompt length or the mere inclusion of strategy labels, but are associated with the specific mapping between learner states and pedagogical strategies.

4.6.6. Human Evaluation of Pedagogical Quality

Since automatic metrics alone cannot fully capture the pedagogical adequacy of tutor responses, we conducted a human evaluation to assess adaptivity and instructional adequacy. The evaluation protocol is explained in

section 4.4.2 in detail. Inter-rater reliability was measured using the intraclass correlation coefficient (ICC), which was 0.66, indicating moderate agreement among annotators.

Adaptivity. Adaptivity measures whether the response takes the student’s level of understanding into account and provides appropriate scaffolding. PATS received an average score of 4.43, compared to 2.98 for the non-PATS baseline and 3.47 for the reference responses. This suggests that annotators perceived PATS responses as more responsive to learner proficiency. The difference between PATS and the baseline was statistically significant ($p < 0.05$).

Teacher-likeness. Teacher-likeness evaluates whether the response resembles a professional teacher’s utterance in tone and instructional style. PATS scored 4.55 on average, compared to 4.18 for non-PATS and 3.53 for the reference responses. This suggests that PATS responses were perceived as more consistent and instructional in tone.

Engagement. Engagement assesses whether the response encourages student participation, for example through follow-up questions or prompts. PATS

Mistral-7B	w/o PATS	43.2 (± 0.3)	0.803 (± 0.002)	0.50 (± 0.01)
	Placebo	43.3 (± 0.3)	0.804 (± 0.002)	0.50 (± 0.01)
	RSS	41.4 (± 0.4)	0.799 (± 0.003)	0.51 (± 0.02)
	PATS	44.6 (± 0.2)[†]	0.815 (± 0.002)[†]	0.54 (± 0.01)[†]
GPT-4	w/o PATS	41.2 (± 0.3)	0.825 (± 0.002)	0.48 (± 0.03)
	Placebo	41.3 (± 0.3)	0.826 (± 0.002)	0.48 (± 0.02)
	RSS	40.7 (± 0.4)	0.823 (± 0.003)	0.49 (± 0.02)
	PATS	42.1 (± 0.2)[†]	0.834 (± 0.001)[†]	0.50 (± 0.01)[†]
Gemini 2.5 Flash	w/o PATS	41.3 (± 0.3)	0.823 (± 0.002)	0.48 (± 0.01)
	Placebo	41.4 (± 0.3)	0.824 (± 0.002)	0.48 (± 0.01)
	RSS	40.9 (± 0.4)	0.821 (± 0.003)	0.49 (± 0.01)
	PATS	42.1 (± 0.2)[†]	0.840 (± 0.001)[†]	0.51 (± 0.01)[†]

Table 9: Prompt-length and strategy-specificity controlled evaluation on the TSCC dataset. Bold denotes the best performance, and [†] indicates statistical significance ($p < 0.05$) compared to the Placebo and RSS variants (Wilcoxon signed-rank test).

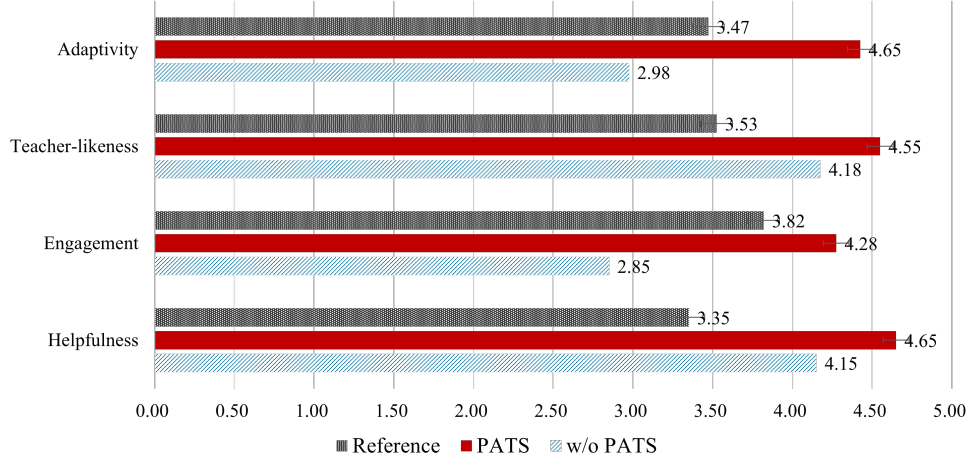


Figure 4: Comparison of PATS, non-PATS, and reference responses across four human evaluation criteria: adaptivity, engagement, teacher-likeness, and helpfulness. Error bars indicate standard deviation.

received a score of 4.28, compared to 2.85 for non-PATS and 3.82 for the reference responses. This indicates that PATS responses more frequently included interactive elements.

Helpfulness. Helpfulness measures the extent to which the response supports the student’s understanding or addresses their difficulty. PATS received an average score of 4.65, compared to 4.15 for non-PATS and 3.35 for the reference responses.

Across all four dimensions, PATS responses were rated higher than the non-PATS baseline in human evaluation. All the difference between PATS and the baseline was statistically significant ($p < 0.05$). The consistent pattern across criteria suggests that integrating pedagogical strategy guidance improves the perceived instructional quality of generated responses.

4.6.7. Qualitative Analysis

Table 10 shows example responses generated by GPT-4 under two conditions: without PATS and with PATS.

In the first interaction, when the student describes “rarely” as a normal word, the baseline model provides a simple clarification but does not promote further engagement. In contrast, the PATS response includes *Scaffolding: Extension*, inviting the student with high mastery and high openness to create

a new sentence using “rarely.” This turns the exchange from a one-way correction into an interactive learning moment that encourages active language use. In the second example, the baseline response offers praise but does not deepen understanding. The PATS version instead uses a *Referential Question*, prompting the student to connect the idiom “rock the boat” to a real-life situation. This helps the student with medium-mastery and high-openness internalize the meaning by linking the expression to personal experience.

These examples show how PATS adapts feedback to individual learner profiles. Students with High Mastery and High Openness benefit from exploratory or reflective questioning, while those with Mid Mastery respond better to prompts that involve production and contextualization.

4.6.8. Error Analysis

Table 11 presents two cases in which PATS generates responses that are accurate in content but misaligned with the local conversational context.

In Case A, the final student turn is a brief procedural exchange about page navigation. Although High Openness inferred from the session history provides a reasonable basis for selecting a Referential question, the immediate turn is a brief procedural exchange about page navigation that carries no instructional content, and the activated strategy is therefore misaligned with the local turn function. The response redirects to a prior vocabulary point rather than following the natural flow of the interaction.

Case B illustrates a different failure mode. Here, the strategy selection

Teacher: OK<STUDENT>one more! -'rarely'=formal or normal?		
Student: I think it's normal		
w/o PATS	[Form-focused feedback]	"I think 'rarely' can be used both formally and informally, but it might be seen as a bit more formal."
PATS	[Form-focused feedback] [Scaffolding: Extension]	" 'rarely' can indeed sound a bit more formal. However, it is also used in normal, everyday speech. Now, let's push a bit further. Can you come up with a sentence using the word 'rarely'?"
Student profile: High Mastery, High Openness, Mid Conscientiousness, Mid Extraversion, High Agreeableness, Low Negative Sensitivity		
(The student is trying to match each phrase with its definition.)		
Student: Hmmmm... you mean you feel confused?		
w/o PATS	-	"Excellent job, <STUDENT>! That's exactly right. You've managed to correctly match each phrase with its definition. Keep up the good work!"
PATS	[Referential Question]	"Well done, <STUDENT>! 'Rock the boat' is used when someone disrupts a situation or causes problems, similar to how literal movement can rock a boat on the water. Now, let's delve a little deeper. Can you provide a real-life situation where you might 'rock the boat'?"
Student profile: Mid Mastery, High Openness, High Conscientiousness, Mid Extraversion, High Agreeableness, Mid Negative Sensitivity		

Table 10: Example interactions with and without PATS with analyzed student traits.

Case	Dialogue Context	Strategy	Generated Response
Case A	<i>Student</i> : “Shall we go to the last test?”	Referential question (high openness)	“Can you recall the key point about the use of ‘it varies’ that we discussed earlier? ...” (procedural turn)
	<i>Teacher</i> : “yes ok”		
	<i>Student</i> : “page 149”		
Case B	<i>Student</i> : “That’s interesting! Are these idioms?”	Scaffolding: Extension (high mastery)	“Let’s extend our learning a bit further. Now, can you tell me what ‘burning bridges’ might mean as an idiom? ...” (over-extension)
	<i>Teacher</i> : “yes they are — but they’re very common ones you know”		
	<i>Student</i> : “4 a sorry”		

Table 11: Representative error cases from the TSCC dataset (GPT-4). Dialogue contexts are excerpted from the last turns, and strategies shown are the primary activated strategy in each case.

(Scaffolding: Extension) is appropriate given the inferred profile (High mastery), and the student has just completed an idiom-matching task successfully. However, the generated response introduces a new idiom (“burning bridges”) beyond the scope of the current exercise, extending the interaction further than the context requires. The content is accurate, but the degree of extension is not warranted by the immediate instructional moment.

Together, these cases suggest that strategy activation based solely on mastery and behavioral tendencies may not capture the full conversational context, particularly the local discourse function of individual turns. As noted in the limitations, incorporating richer contextual signals (e.g. interaction phase or task completion state) could further refine strategy selection in future work.

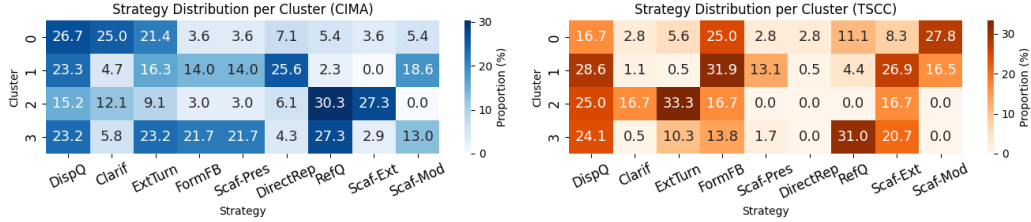
4.6.9. Differential Effects Across Learner Profiles

We conducted a clustering analysis to examine how inferred student traits relate to the teaching strategies generated by the model and to the resulting performance. The results are summarized in Figure 5a for learner profiles and performance, Figure 5b and Figure 5c for strategy distribution. K-means clustering ($k = 4$) was applied separately to CIMA and TSCC because the two datasets differ in dialogue length and trait distributions.

Figure 5a illustrates the distinct learner profiles that emerged from our analysis of mastery and behavioral tendencies. In the CIMA dataset, we ob-

Dataset	Cluster	Sample size	Mastery	Openness	Conscientiousness	Extraversion	Agreeableness	Negative sensitivity	BERTScore F1	DialogRPT
CIMA	0	112	0.84	1.68	1.16	0.16	0.98	0.12	0.8185	0.5273
	1	86	0.12	1.96	0.19	0.88	1.49	0.33	0.8282	0.5439
	2	66	1.92	1.88	1.36	0.48	1.60	0.01	0.8360	0.4834
	3	138	0.87	1.78	1.09	0.19	1.88	0.12	0.8193	0.5217
TSCC	0	36	1.17	1.94	1	1.88	1.97	0	0.8312	0.4882
	1	182	1.11	1.9	1.57	1.22	1.73	0.04	0.8282	0.4910
	2	12	1	1.5	1.33	0.11	1.63	0	0.8242	0.5080
	3	58	2	1.92	1.87	1.4	1.83	0	0.8335	0.4384

(a) Cluster-level averages of student traits (0=Low, 1=Mid, 2=High) and model performance (BERTScore F1 and DialogRPT).



(b) CIMA strategy distribution.

(c) TSCC strategy distribution.

Figure 5: Clustering analysis of trait-strategy-performance patterns.

serve differences in proficiency and diligence across the groups. Cluster 2 is characterized by high-mastery learners ($M = 1.92$) who also show moderate-to-high conscientiousness. By contrast, Cluster 1 contains low-mastery learners ($M = 0.12$) with high openness scores but notably low conscientiousness ($C = 0.19$). The remaining groups, Clusters 0 and 3, comprise learners with mid-level mastery. Cluster 3 stands out due to its high agreeableness score ($A = 1.88$), indicating a particularly cooperative learner profile. The TSCC dataset also revealed distinct groups, primarily defined by their social and behavioral traits. Cluster 3 represents the most proficient learners ($M = 2.00$), who also display high conscientiousness. Cluster 2 shows an introverted profile, marked by the lowest extraversion score ($E = 0.11$). Conversely, Cluster 0 shows the highest extraversion ($E = 1.88$) combined with relatively lower conscientiousness ($C = 1.00$). Finally, Cluster 1 is the largest subgroup ($N = 182$). These are mid-mastery learners with high openness.

Performance metrics varied across different profiles. In CIMA, the high-mastery Cluster 2 achieved the highest BERTScore (0.8360), indicating semantic alignment with reference responses. In TSCC, the introverted Cluster 2 recorded the highest DialogRPT (0.5080). This can indicate that the sys-

tem’s adaptive strategies effectively maintained engagement even with less outgoing learners.

The distribution of teaching strategies is visualized in the heatmap in Figure 5b and Figure 5c. In CIMA, Cluster 2 (High Mastery) frequently utilized higher-level strategies such as *Referential Questions* (RefQ, 30.3%) and *Scaffolding: Extension* (Scaf-Ext, 27.3%). While this yielded the highest BERTScore, the DialogRPT was relatively lower, possibly because complex inquiries are less favored by engagement-oriented metrics. Conversely, CIMA Cluster 1 (Low Mastery) primarily received support through *Direct Repair* (DirectRep, 25.6%) and *Scaffolding: Modeling* (Scaf-Mod, 18.6%). Interestingly, this group exhibited the highest DialogRPT scores (0.5439) within CIMA, indicating that directive scaffolding is effective for engaging struggling learners. In TSCC, Cluster 2 (Low Extraversion) showed a preference for *Extended Teacher Turn* (ExtTurn, 33.3%). This aligns with pedagogical theory suggesting that introverted students often benefit from teacher-led explanations that reduce the pressure for immediate social interaction. Thus, from the cluster analysis, it is observed that the generated teaching strategies vary according to the combinations of learner traits.

5. Ethical Considerations

Inferring personality-related traits from dialogue raises concerns regarding privacy, profiling, and fairness. Unlike short-term performance indicators, personality signals are relatively stable and may lead to unintended differential treatment. In this study, trait information is used solely for experimental purposes and is not linked to identifiable personal data. The system does not store persistent personality profiles beyond the interaction context.

For real-world deployment, explicit user consent and transparent disclosure of the adaptive mechanism are required, along with clear opt-out options. Data collection should be limited to information strictly necessary for instructional support. Trait-based adaptation must be continuously monitored to prevent unintended disadvantages for specific learner profiles. In particular, systematic subgroup-level evaluations are necessary to detect potential disparities in tone, feedback style, or instructional quality across trait clusters. Due to the controlled experimental scope of the present study, such large-scale subgroup analyses were beyond scope. Future work will extend the evaluation to include comprehensive bias assessments across learner groups. When inference confidence is low, the system should default to less direc-

tive strategies rather than reinforcing potentially inaccurate profiles. More broadly, trait-based adaptation should function as supportive guidance rather than prescriptive categorization, and high-stakes decisions should remain under human oversight.

6. Conclusion

In this study, we proposed PATS, a framework for pedagogically adaptive tutoring that uses a student’s behavioral tendencies and mastery level to guide strategy selection during response generation. By incorporating these pedagogically relevant signals into the prompting process, the framework produces more context-sensitive tutor responses. Our experiments show consistent improvements across evaluation metrics, particularly on the TSCC dataset, indicating that structured strategy guidance enhances the instructional quality of generated responses. These gains align with established tutoring principles: strategies such as scaffolding, reasoning prompts, and avoiding direct answers support deeper learning, while trait-aware modulation adjusts their expression to match individual learner profiles.

However, several limitations remain. First, the rule-based engine relies solely on mastery and behavioral tendencies; incorporating richer contextual signals such as lesson stage, error type, or instructional goals could further refine strategy selection. The fixed rule mappings also encode assumptions that may not generalize across domains, age groups, or cultural contexts, and systematic evaluation of domain transfer remains future work. Second, trait inference from short dialogue histories has inherent limitations. Brief exchanges may not fully capture stable traits, trait misclassification can propagate to downstream strategy selection, and the current deterministic routing does not model uncertainty. The student modeling module currently relies on prompt-based inference, as in-domain labeled data for trait annotation is not available; probabilistic routing mechanisms, multimodal signals, or PEFT-based adaptation are directions for future work. Third, since this study focuses on strategy-level alignment, systematic hallucination analysis and safety guardrails are left for future research. Controlled user studies with pre- and post-tests are needed to assess the framework’s impact on knowledge retention and skill development.

The implementation code and experimental settings of PATS are available at: <https://github.com/hjeonn/pats-adaptive-its>

Declaration of Generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author(s) used Google Gemini and Grammarly in order to enhance the linguistic clarity of the manuscript. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

References

- Adigwe, A., Yuan, Z., 2023. The adaio system at the bea-2023 shared task on generating ai teacher responses in educational dialogues. arXiv preprint arXiv:2306.05360 .
- Anderson, J.R., 1983. The architecture of cognition. Psychology Press.
- Arpaci, I., Kuşci, I., Gibreel, O., 2025. The role of personality traits in predicting educational use of generative ai in higher education. Scientific Reports 15, 30440.
- Baradeyah, N., Farrah, M., Hebron, P., 2017. The impact of using asking for clarification and circumlocution speaking strategies on enhancing the speaking skill of efl students. Studies in Linguistics and Literature 1.
- Bibauw, S., François, T., Desmet, P., 2022. Dialogue systems for language learning: Chatbots and beyond, in: The Routledge handbook of second language acquisition and technology. Routledge, pp. 121–135.
- Bidjerano, T., Dai, D.Y., 2007. The relationship between the big-five model of personality and self-regulated learning strategies. Learning and individual differences 17, 69–81.
- Bloom, B.S., 1968. Learning for mastery. instruction and curriculum. regional education laboratory for the carolinas and virginia, topical papers and reprints, number 1. Evaluation comment 1, n2.
- Borchers, C., Deininger, H., Pardos, Z.A., 2026. Toward trait-aware learning analytics. arXiv preprint arXiv:2602.00018 .
- Borchers, C., Shou, T., 2025. Can large language models match tutoring system adaptivity? a benchmarking study, in: International Conference on Artificial Intelligence in Education, Springer. pp. 407–420.

- Brock, C.A., 1986. The effects of referential questions on esl classroom discourse. *TESOL quarterly* 20, 47–59.
- Caines, A., Yannakoudakis, H., Allen, H., Pérez-Paredes, P., Byrne, B., Buttery, P., 2022. The teacher-student chatroom corpus version 2: more lessons, new annotation, automatic detection of sequence shifts, in: *Swedish Language Technology Conference and NLP4CALL*, pp. 23–35.
- Chaiklin, S., et al., 2003. The zone of proximal development in vygotsky’s analysis of learning and instruction. *Vygotsky’s educational theory in cultural context* 1, 39–64.
- Dennis, M., Masthoff, J., Mellish, C., 2016. Adapting progress feedback and emotional support to learner personality. *International Journal of Artificial Intelligence in Education* 26, 877–931.
- Dettmers, T., Pagnoni, A., Holtzman, A., Zettlemoyer, L., 2023. Qlora: Efficient finetuning of quantized llms. *Advances in neural information processing systems* 36, 10088–10115.
- Ellis, R., 1997. *Second language acquisition*. The United States: Oxford 98, 37.
- Fotos, S., Nassaji, H., 2013. *Form-focused instruction and teacher education*. Oxford University Press.
- Gaeta, A., Orcioli, F., Pascuzzo, A., Peduto, A., 2025. Enhancing traditional its architectures with large language models for generating motivational feedback. *Computers and Education: Artificial Intelligence* , 100433.
- Ganesh, A., Palmer, M., Kann, K., 2021. What would a teacher do? predicting future talk moves. *arXiv preprint arXiv:2106.05249* .
- Gao, X., Zhang, Y., Galley, M., Brockett, C., Dolan, B., 2020. Dialogue response ranking training with large-scale human feedback data. *arXiv preprint arXiv:2009.06978* .
- Ghefaili, A., 2003. Cognitive apprenticeship, technology, and the contextualization of learning environments. *Journal of Educational Computing, Design & Online Learning* 4, 1–27.

- Goldberg, L.R., 1990. An alternative" description of personality": The big-five factor structure. *Journal of Personality and Social Psychology* 59, 1216.
- Graesser, A.C., Lu, S., Jackson, G.T., Mitchell, H.H., Ventura, M., Olney, A., Louwerse, M.M., 2004. Autotutor: A tutor with dialogue in natural language. *Behavior Research Methods, Instruments, & Computers* 36, 180–192.
- Grattafiori, A., Dubey, A., Jauhri, A., Pandey, A., Kadian, A., Al-Dahle, A., Letman, A., Mathur, A., Schelten, A., Vaughan, A., et al., 2024. The llama 3 herd of models. *arXiv preprint arXiv:2407.21783* .
- Hattie, J., Timperley, H., 2007. The power of feedback. *Review of educational research* 77, 81–112.
- Hicke, Y., Masand, A., Guo, W., Gangavarapu, T., 2023. Assessing the efficacy of large language models in generating accurate teacher responses. *arXiv preprint arXiv:2307.04274* .
- Hu, E.J., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S., Wang, L., Chen, W., 2021. Lora: Low-rank adaptation of large language models. *arXiv preprint arXiv:2106.09685* .
- Huber, T., Niklaus, C., Handschuh, S., 2023. Enhancing educational dialogues: A reinforcement learning approach for generating ai teacher responses, in: *Proceedings of the 18th Workshop on Innovative Use of NLP for Building Educational Applications (BEA 2023)*, pp. 736–744.
- Ikram, F., Scarlatos, A., Lan, A., 2025. Exploring llms for predicting tutor strategy and student outcomes in dialogues. *arXiv preprint arXiv:2507.06910* .
- Jensen, M., 2015. Personality traits, learning and academic achievements. *Journal of Education and Learning* 4, 91–118.
- Jiang, A.Q., Sablayrolles, A., Mensch, A., Bamford, C., Chaplot, D.S., de las Casas, D., Bressand, F., Lengyel, G., Lample, G., Saulnier, L., Lavaud, L.R., Lachaux, M.A., Stock, P., Scao, T.L., Lavril, T., Wang, T., Lacroix, T., Sayed, W.E., 2023. Mistral 7b. URL: <https://arxiv.org/abs/2310.06825>, *arXiv:2310.06825*.

- Katz, S., Albacete, P., Chounta, I.A., Jordan, P., McLaren, B.M., Zapata-Rivera, D., 2021. Linking dialogue with student modelling to create an adaptive tutoring system for conceptual physics. *International journal of artificial intelligence in education* 31, 397–445.
- KhanAcademy, 2024. Meet khanmigo, khan academy’s ai-powered teaching assistant tutor. Internet. Available from: <https://www.khanmigo.ai/> .
- Komarraju, M., Karau, S.J., Schmeck, R.R., Avdic, A., 2011. The big five personality traits, learning styles, and academic achievement. *Personality and individual differences* 51, 472–477.
- Landis, J.R., Koch, G.G., 1977. The measurement of observer agreement for categorical data. *biometrics* , 159–174.
- Lin, C.Y., 2004. Rouge: A package for automatic evaluation of summaries, in: *Text summarization branches out*, pp. 74–81.
- Lin, J., Tan, W., Du, L., Buntine, W., Lang, D., Gašević, D., Chen, G., 2023. Enhancing educational dialogue act classification with discourse context and sample informativeness. *IEEE Transactions on Learning Technologies* .
- Lin, Y.H., Hedgcock, J., 1996. Negative feedback incorporation among high-proficiency and low-proficiency chinese-speaking learners of spanish. *Language Learning* 46, 567–611.
- Liu, C., Hoang, L., Stolman, A., Kizilcec, R.F., Wu, B., 2025. Understanding student engagement with large language model-powered course assistants, in: *International Conference on Artificial Intelligence in Education*, Springer. pp. 3–10.
- Lu, Y., Wang, D., Chen, P., Zhang, Z., 2024. Design and evaluation of trustworthy knowledge tracing model for intelligent tutoring system. *IEEE Transactions on Learning Technologies* .
- Macina, J., Daheim, N., Wang, L., Sinha, T., Kapur, M., Gurevych, I., Sachan, M., 2023. Opportunities and challenges in neural dialog tutoring. *arXiv preprint arXiv:2301.09919* .

- Mammadov, S., 2022. Big five personality traits and academic performance: A meta-analysis. *Journal of personality* 90, 222–255.
- Omidvar, A., An, A., 2023. Empowering conversational agents using semantic in-context learning, in: *Proceedings of the 18th Workshop on Innovative Use of NLP for Building Educational Applications (BEA 2023)*, pp. 766–771.
- Piro, L., Bianchi, T., Alessandrelli, L., Chizzola, A., Casiraghi, D., Sancasani, S., Gatti, N., 2024. Mylearningtalk: An llm-based intelligent tutoring system, in: *International Conference on Web Engineering*, Springer. pp. 428–431.
- Poropat, A.E., 2009. A meta-analysis of the five-factor model of personality and academic performance. *Psychological bulletin* 135, 322.
- Raffel, C., Shazeer, N., Roberts, A., Lee, K., Narang, S., Matena, M., Zhou, Y., Li, W., Liu, P.J., 2020. Exploring the limits of transfer learning with a unified text-to-text transformer. *Journal of machine learning research* 21, 1–67.
- Rani Bhattacharjee, R., Ramkumar, A., 2025. Effect of big five personality dimensions on the academic performance of college students. *Frontiers in Psychology* 16, 1490427.
- Razafinirina, M.A., Dimbisoa, W.G., Mahatody, T., 2024. Pedagogical alignment of large language models (llm) for personalized learning: A survey, trends and challenges. *Journal of Intelligent Learning Systems and Applications* 16, 448–480.
- Scarlatos, A., Liu, N., Lee, J., Baraniuk, R., Lan, A., 2025. Training llm-based tutors to improve student learning outcomes in dialogues, in: *International Conference on Artificial Intelligence in Education*, Springer. pp. 251–266.
- Schmucker, R., Xia, M., Azaria, A., Mitchell, T., 2024. Ruffle&riley: Insights from designing and evaluating a large language model-based conversational tutoring system, in: *International Conference on Artificial Intelligence in Education*, Springer. pp. 75–90.
- Shomoossi, N., 1997. The effect of teacher’s questioning behavior on eff classroom interaction: A classroom-based research. *Online Submission* .

- Sonkar, S., Ni, K., Chaudhary, S., Baraniuk, R.G., 2024. Pedagogical alignment of large language models. arXiv preprint arXiv:2402.05000 .
- Sottolare, R.A., Graesser, A., Hu, X., Holden, H., 2013. Design recommendations for intelligent tutoring systems: Volume 1-learner modeling. volume 1. US Army Research Laboratory.
- Stasaski, K., Kao, K., Hearst, M.A., 2020. Cima: A large open access dialogue dataset for tutoring, in: Proceedings of the Fifteenth Workshop on Innovative Use of NLP for Building Educational Applications, pp. 52–64.
- Suresh, A., Sumner, T., Jacobs, J., Foland, B., Ward, W., 2019. Automating analysis and feedback to improve mathematics teachers’ classroom discourse, in: Proceedings of the AAAI conference on artificial intelligence, pp. 9721–9728.
- Swain, M., 1995. Three functions of output in second language learning. Principles and practice in applied linguistics: Studies in honor of HG Widdowson , 125–144.
- Trown, E.A., 1970. Some evidence on the interaction between teaching strategy and personality. British Journal of Educational Psychology 40, 209–211.
- Upadhyay, S., Ginsberg, E., Callison-Burch, C., 2023. Improving mathematics tutoring with a code scratchpad, in: Proceedings of the 18th Workshop on Innovative Use of NLP for Building Educational Applications (BEA 2023), pp. 20–28.
- Vasselli, J., Vasselli, C., Nohejl, A., Watanabe, T., 2023. Naisteacher: A prompt and rerank approach to generating teacher utterances in educational dialogues, in: Proceedings of the 18th Workshop on Innovative Use of NLP for Building Educational Applications (BEA 2023), pp. 772–784.
- Vygotsky, L.S., Cole, M., 1978. Mind in society: Development of higher psychological processes. Harvard university press.
- Walsh, S., 2003. Developing interactional awareness in the second language classroom through teacher self-evaluation. Language awareness 12, 124–142.

- Walsh, S., 2006. Investigating classroom discourse. Routledge.
- Wang, L., Sachan, M., Zeng, X., Wong, K.F., 2023. Strategize before teaching: A conversational tutoring system with pedagogy self-distillation, in: Vlachos, A., Augenstein, I. (Eds.), Findings of the Association for Computational Linguistics: EACL 2023, Association for Computational Linguistics, Dubrovnik, Croatia. pp. 2268–2274. URL: <https://aclanthology.org/2023.findings-eacl.170/>, doi:10.18653/v1/2023.findings-eacl.170.
- Zhang, L., Lin, J., Kuang, Z., Xu, S., Hu, X., 2024. Spl: a socratic playground for learning powered by large language model. arXiv preprint arXiv:2406.13919 .
- Zhang, T., Kishore, V., Wu, F., Weinberger, K.Q., Artzi, Y., 2019. Bertscore: Evaluating text generation with bert. arXiv preprint arXiv:1904.09675 .

Appendix A. Appendix

Appendix A.1. Trait Inference Prompt (P_{trait})

The trait inference prompt assigns the model the role of an educational psychologist and requires categorical outputs for mastery and Big Five traits.

Instructional context.

Based on the following conversation, predict the student’s mastery level and Big Five traits as **High**, **Mid**, or **Low**. Use the definitions below.

Openness

[High] Creativity in answers; open to new ideas or challenges.

[Mid] Occasionally curious or flexible.

[Low] Prefers routine responses; avoids new challenges.

Conscientiousness

[High] Careful and organized; attentive to detail.

[Mid] Moderately consistent and responsible.

[Low] Careless or inconsistent.

Extraversion

[High] Actively engages and expresses ideas openly.

[Mid] Participates when prompted.

[Low] Brief responses; limited engagement.

(Definitions for Agreeableness and Negative Sensitivity follow the same format.)

Output constraints.

Return the result in strict JSON format:

```
{
  "Mastery": "",
  "Big Five": {
    "Openness": "",
    "Conscientiousness": "",
    "Extraversion": "",
    "Agreeableness": "",
    "Negative Sensitivity": ""
  }
}
```

Dialogue history (example).

Teacher: “black” is “blu” in Italian.

Student: But what are the other words?